



Cyprus
University of
Technology

CEI 327 Παράλληλη Επεξεργασία και Κατανεμημένα Συστήματα

Εαρινό Εξάμηνο 2024

Τμήμα Ηλεκτρολόγων Μηχανικών και Μηχανικών Ηλεκτρονικών Υπολογιστών
και Πληροφορικής
Τεχνολογικό Πανεπιστήμιο Κύπρου

Τελική Εξέταση (45%)

16/5/2024, 17:00-20:00

Ονοματεπώνυμο:

Αριθμός Φοιτητικής Ταυτότητας:

Επιτρέπονται ανοικτές σημειώσεις, λάπτοπ και Internet. Δεν επιτρέπεται χρήση εργαλείων τύπου GPT και μηχανών αναζήτησης και η χρήση κινητού τηλεφώνου. Οποιαδήποτε μορφή αντιγραφής, παράτυπης συνεργασίας και ηλεκτρονικής επικοινωνίας με συμφοιτητές σας την ώρα της εξέτασης συνεπάγεται αυτοπερθέτως την αποτυχία στο μάθημα, ανεξαρτήτως της βαθμολογίας στις υπόλοιπες πτυχές του μαθήματος.

ΤΕΧΝΟΛΟΓΙΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ
ΚΥΠΡΟΥ

Question 1 (6%)

- Explain the concept of a layered architecture in distributed systems.
- Provide a real-world example of a distributed system that uses a layered architecture, describing each layer's role.
- Compare and contrast object-based and service-oriented architectural styles.
- Discuss the benefits and drawbacks of using a RESTful architecture.
- How does the publish-subscribe architecture differ from the client-server model in terms of communication and synchronization?

Question 2 (6%)

- a. Differentiate between a thread and a process in the context of distributed systems.
- b. Explain the advantages of using multithreading in a server application.
- c. Describe how context switching works between threads within the same process.
- d. Consider a web server handling multiple client requests. How would you implement a multithreaded model to improve performance? Provide a high-level design.
- e. Discuss the potential issues of multithreading and how you would mitigate one of these issues.

Question 3 (8%)

- a. Define the Remote Procedure Call (RPC) and describe its basic operation.
- b. Explain how parameter passing works in RPC and the challenges associated with it.
- c. Differentiate between synchronous RPC and asynchronous RPC with examples.
- d. Discuss the role of stubs in RPC. How do they contribute to the transparency of the procedure call?
- e. Consider a situation where a client and server have different data representations (e.g., different byte ordering). How does parameter passing address this issue? Provide a concrete example.
- f. Explain the concept of remote references in RPC and how they enhance access transparency.

Question 4 (6%)

1. Give a high level description of the Chord protocol and its organization of nodes in a logical ring. How are nodes assigned identifiers, and what is the role of the successor in relation to entity keys?
2. Explain the concept of finger tables in the Chord protocol. What information do finger tables contain, and how do they facilitate key lookups in the system?
3. Discuss the challenges posed by network proximity in the Chord protocol. How can erratic and long-delay message transfers be mitigated? Describe the solutions of topology-aware node assignment and proximity routing.

Question 5 (6%)

- Explain the roles of TCP and UDP in the transport layer. Provide examples of applications suited for each protocol.
- Discuss the concept of middleware protocols and their importance in distributed systems.
- Compare transient and persistent communication, providing examples for each.
- Describe the role of sockets in message-oriented communication and provide a simple Python example of a client-server interaction using sockets.

Question 6 (6%)

- Define the concept of naming in distributed systems and its importance.
- Explain the difference between flat naming and structured naming.
- Describe the DNS system and how it helps in resolving names in distributed systems.
- Discuss the challenges of naming in peer-to-peer systems and suggest possible solutions.
- Illustrate with an example how naming can affect the performance of a distributed system.

Question 7 (7%)

- Define total ordered multicast and explain its significance in distributed systems.
- Describe an algorithm to achieve total ordered multicast using Lamport timestamps.
- Consider a scenario where three processes P1, P2, and P3 need to send messages m1, m2, and m3, respectively. Using Lamport timestamps, illustrate the order in which these messages will be delivered if P1 sends m1 at time 2, P2 sends m2 at time 1, and P3 sends m3 at time 3.
- Identify potential challenges when implementing total ordered multicast in a distributed system.
- Suggest ways to address one of the challenges identified in part (d).